DO GROUPS MAKE WISE DECISIONS?

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Motivation

- Hiring committees, mutual funds, etc. make decisions jointly
- Joint decision-making comes with **benefits** but also **risks:**
  - *Benefits*: diverse inputs, deliberation, weighing best inputs
  - *Risks*: conflicting inputs, herding, overconfident teammates

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$\Rightarrow$ Joint and individual decision-making differ in **several respects at once**

$\Rightarrow$ Which features of joint decision-making **enhance decision quality**, and which **harm it**?

Distinct features of joint decision-making

Joint decision: a shared decision made by a group after discussion

1. **Composition**

2. **Communication**

3. **Collaborative aggregation**

Distinct features of joint decision-making

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1. **Composition**
   - Number and mix of people making a decision (+ inputs they bring)
2. **Communication**
   - Exchange and discussion of inputs
3. **Collaborative aggregation**
   - Agreeing on joint decision rather than aggregating separate judgments

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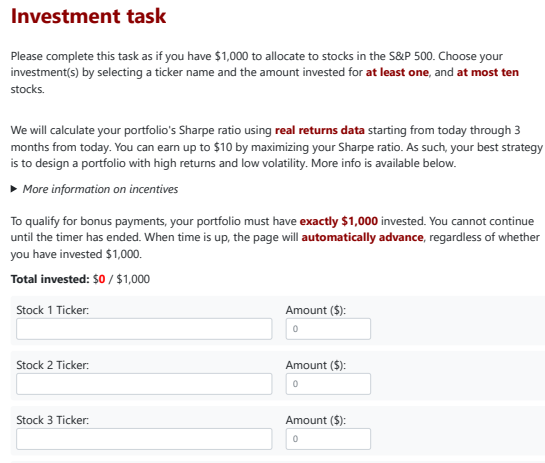
$\Rightarrow$ Need **comparable decisions** & **exogenous variation** in each feature

$\Rightarrow$ Experiment to estimate the **separate contribution** of each one

Design

Main task: portfolio allocation

- Hypothetical $1,000 to invest
  - Searchable list of S&P 500 stocks
- Maximize the **Sharpe ratio**
  - Increase returns, reduce risk
  - Bonus up to $10 based on Sharpe
- Randomly assigned **decision mode**



Decision mode visualization

INDIVIDUALS	CROWDS	PANELS	TEAMS
			
Own portfolio	**Pooled portfolio**	Pooled portfolio	**Joint portfolio**
Up to 10 stocks	**Up to 20 stocks**	Up to 20 stocks	Up to 20 stocks
No communication	No communication	**Communication**	Communication
128 individuals	64 pairs	64 pairs	66 pairs

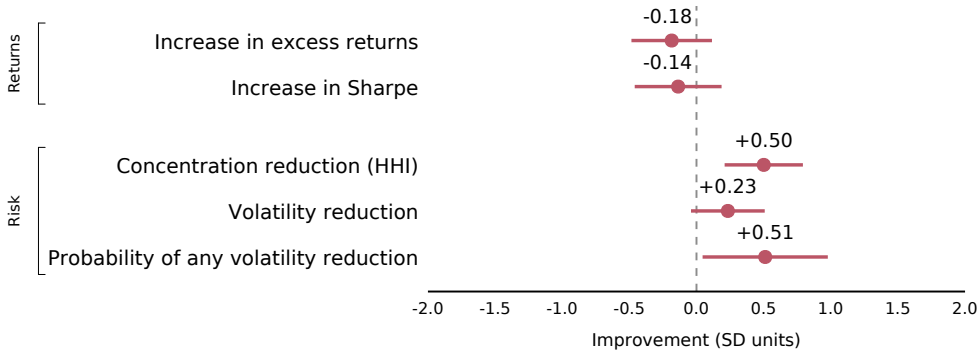
Portfolio aggregation

- *Crowd* and *Panel* aggregation via portfolio **pooling**
  - Each partner submits a separate portfolio, e.g.:
  - Partner A: $100 in Stock A, Partner B: $100 in Stock B
  - Pooled portfolio: $50 in Stock A, $50 in Stock B
- *Teams* aggregate **collaboratively**
  - Shared portfolio input fields, both teammates must agree
  - Not restricted to 50-50 allocation

Results

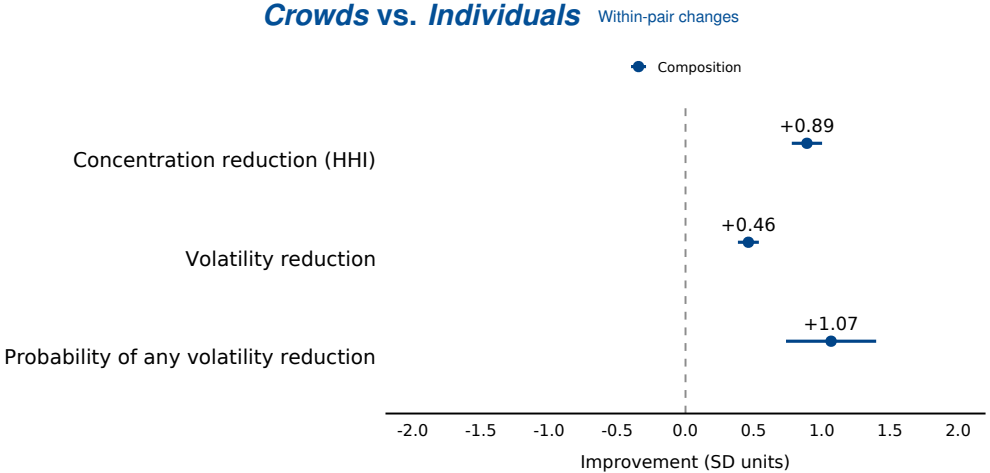
Teams outperform *Individuals* on portfolio risk

Teams* vs. *Individuals, controlling for baseline performance Within-pair changes



- But **which features** of joint decision-making explain effects on portfolio risk?

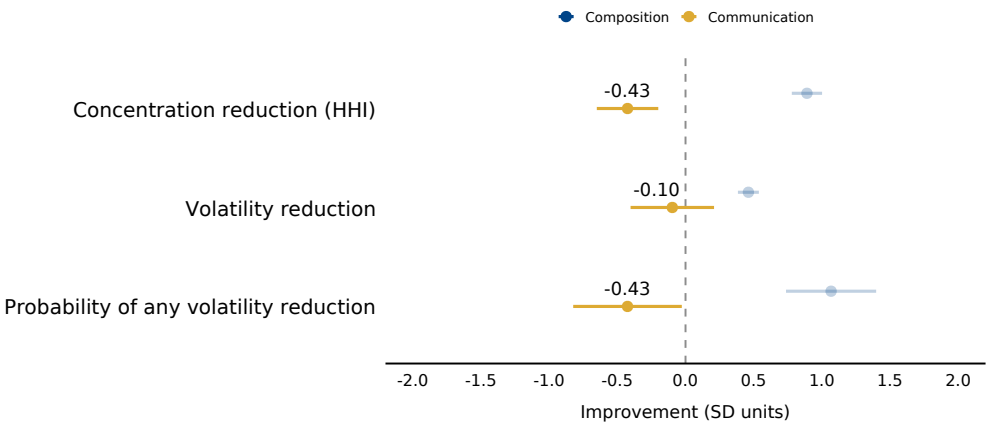
Strong reductions in risk from combining inputs



- Compositional effects **exceed total effect** of joint decision-making

Communication undermines diversification

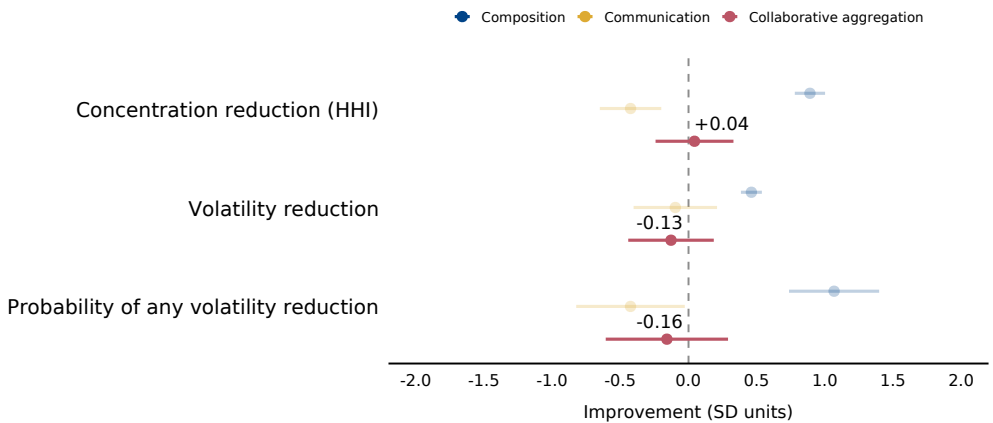
Panels* vs. *Crowds, controlling for baseline performance Within-pair changes



- Communication **attenuates the gains** from combining inputs

No average effect of collaborative aggregation

Teams* vs. *Panels, controlling for baseline performance Within-pair changes



- **Small** and **inconsistent** effects on portfolio risk

Mechanisms & Heterogeneity

Understanding our results

Mechanisms

- Why does pooling inputs reduce portfolio risk?
  - Diversification & lower exposure to idiosyncratic asset risk
  - Analogous to **wisdom-of-crowd** effects observed in prior tasks

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  - H1: it undermines diversification by making portfolios **more similar**
  - H2: it **shifts attention** away from diversification toward returns

Understanding our results

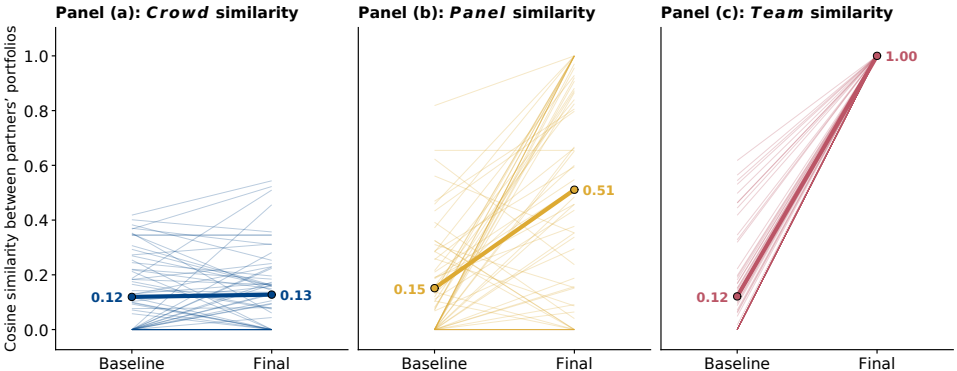
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Heterogeneity

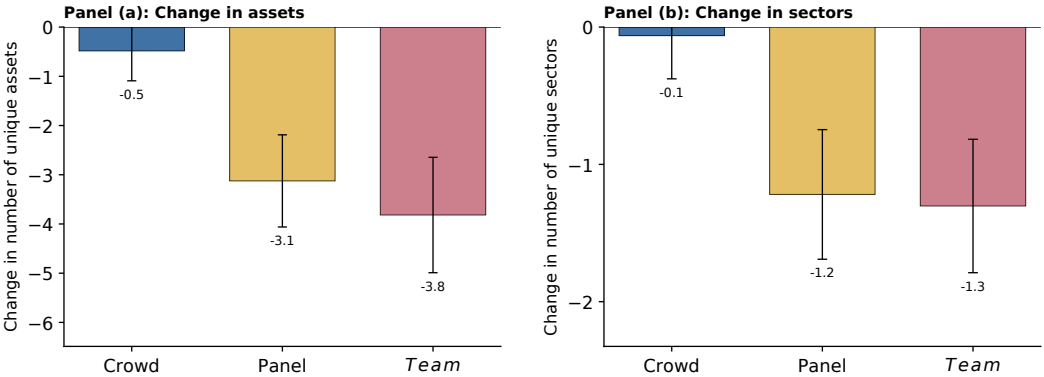
- Do these effects depend on **who is paired** together?
- **Cognitive diversity**, i.e. differences in beliefs & other inputs, may matter

Communication makes portfolios more similar



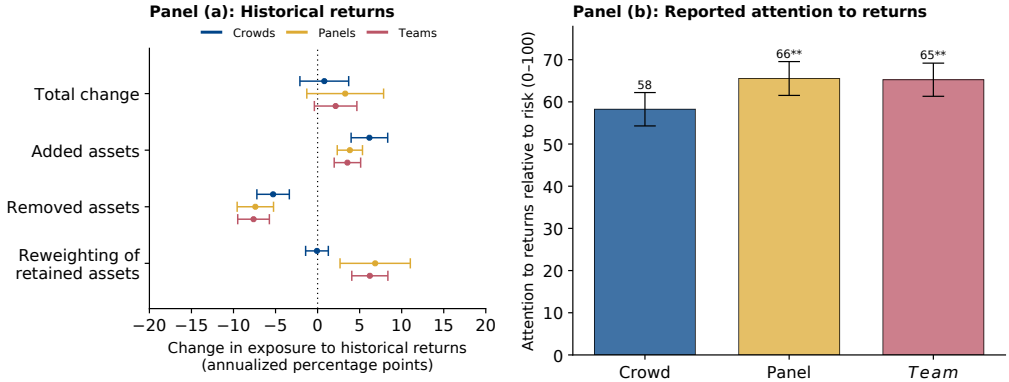
- *Panel* member submissions **converge** despite making **separate decisions**

Communication reduces portfolio breadth



- *Panels* & *Teams* **drop unique assets & sectors** held in baseline portfolios

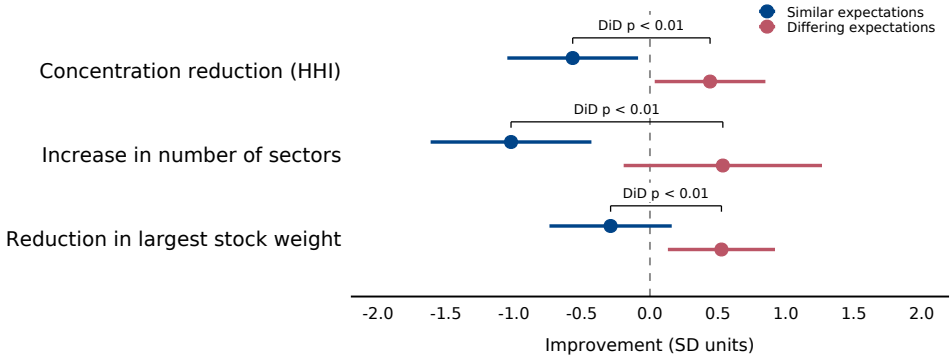
Communication shifts focus to returns



- Members **double down** on stocks with high historical returns held at baseline
- Higher reported **focus on returns** in communicating modes (both $p < 0.05$)

Aggregation effects by belief diversity

Teams* vs. *Panels, controlling for baseline performance Within-pair changes



- Collaborative aggregation improves diversification for **belief-diverse** pairs

Alternative explanations

Are these results:

1. An inevitable consequence of our design? **No**
   - Subjects know that diversification is a good strategy Evidence
   - Pooling aids diversification & is easy for *Teams* to reproduce
   - *Teams* have larger toolkit: communication and collaborative aggregation

2. Driven by copying of LLM portfolios? **No**
   - Human and LLM portfolios differ systematically Stock choice Breadth

3. Dependent on the evaluation window? **No**
   - Strongest effects in window-invariant measures, e.g. concentration
   - Similar results in alternative evaluation window Evidence

Takeaways

- *Crowds*, not *Teams*, perform best on portfolio risk
- *Panel* & *Team* underperformance driven mostly by **communication**
  - Attenuates gains from pooling
  - Shifts focus to returns, but fails to increase them
- Collaborative aggregation can help when groups hold **differing expectations**

Takeaways

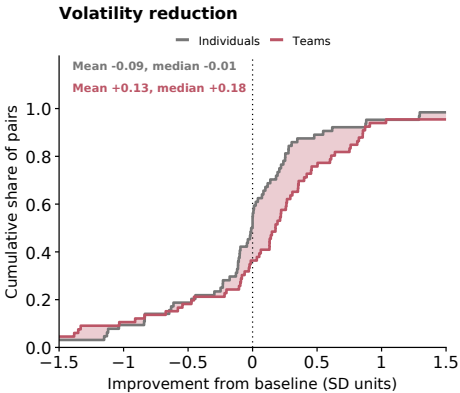
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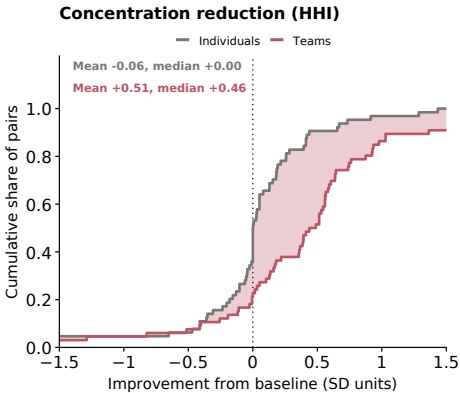
$\Rightarrow$ Decomposition reveals **offsetting effects** that naive comparisons would miss

$\Rightarrow$ **Groups with access to diverse inputs may not fully utilize them**

Appendix

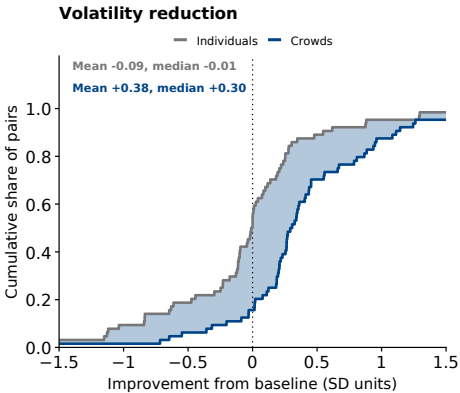
Within-pair changes: *Teams* and *Individuals*

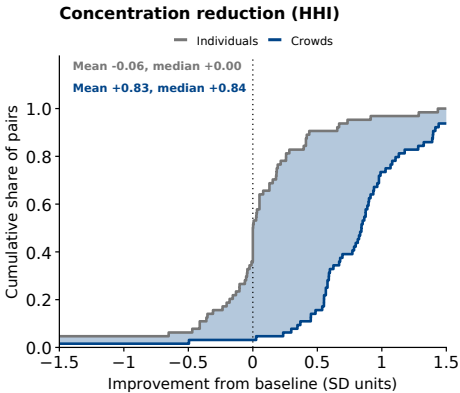




◂ Back to coefficients

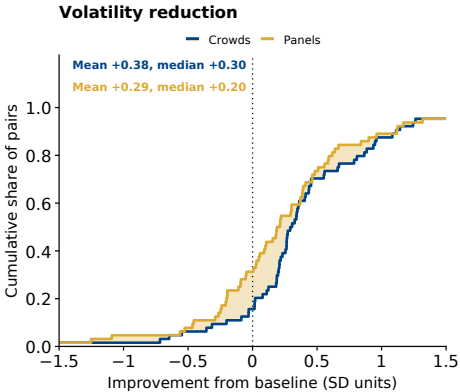
Within-pair changes: Crowds and *Individuals*

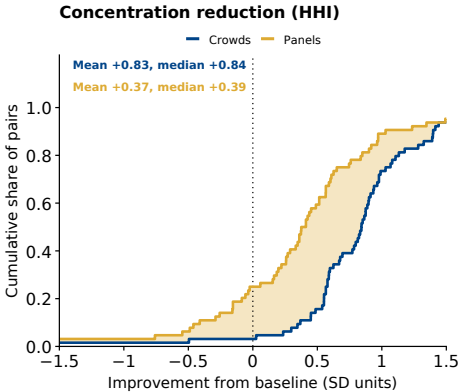




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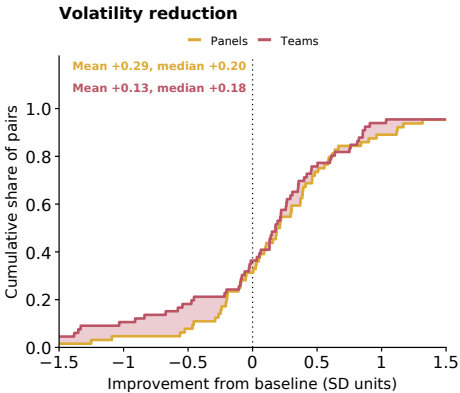
Within-pair changes: Panels and Crowds

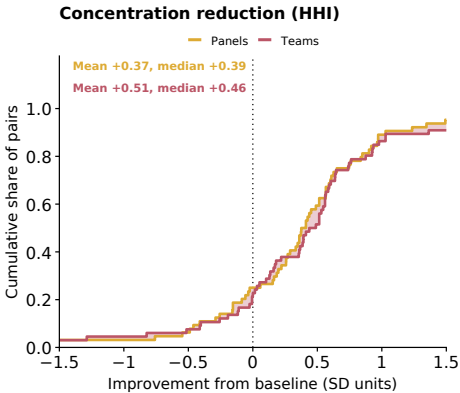




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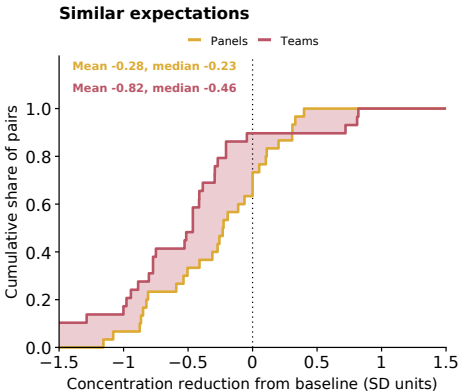
Within-pair changes: *Teams* and Panels

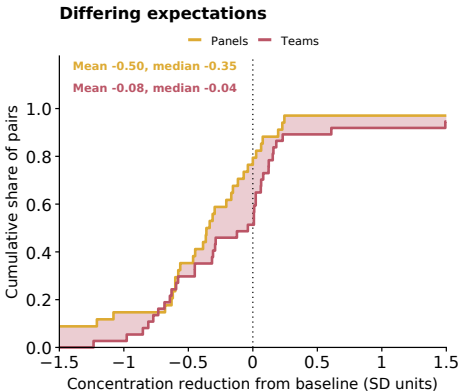




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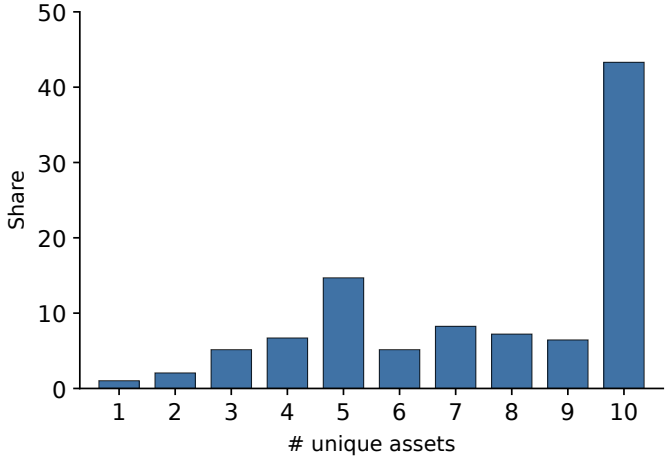
Within-pair changes by expectations





◂ Back to coefficients

Individuals spread across assets

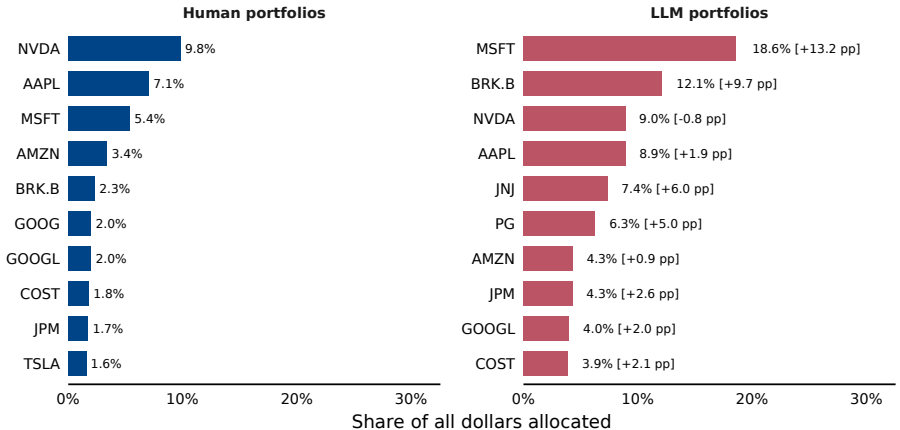


- Mean **7.6** unique assets
- **43%** allocate the maximum of 10

⇒ Attempts to **diversify** in line with incentives

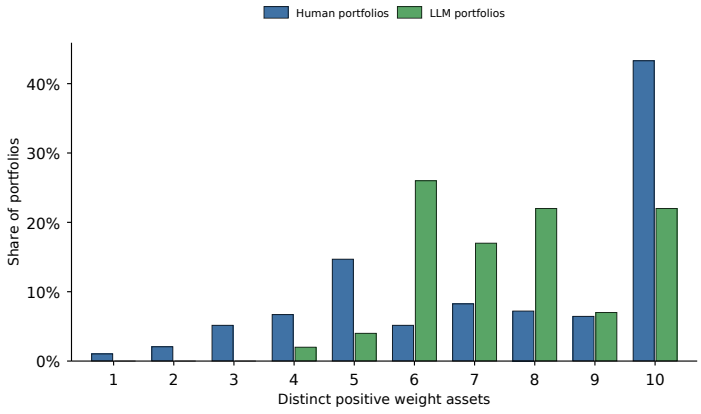
◂ Alternative explanations

Human and LLM stock choices



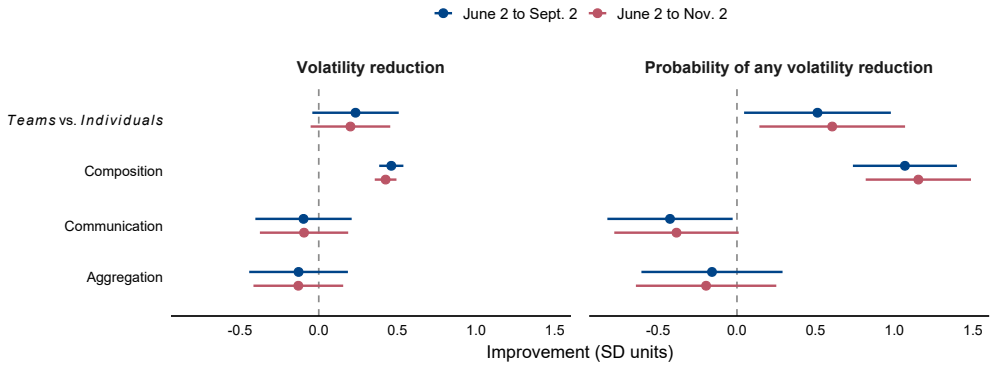
◂ Alternative explanations

Human and LLM portfolio breadth



Alternative explanations

Alternative evaluation window



◂ Alternative explanations